Keep the grid stable

Dynamic topology optimization & converter tuning

TenSyGrid 1-day hackathon challenge

Universitat Politècnica de Catalunya · CETPartnership Horizon Europe (GA 101069750)

Power systems are rapidly transitioning towards inverter-dominated operation. This 1-day hackathon challenges teams to first **reconstruct a function from structural and coefficient matrices** in an ungraded one-hour warm-up, then use the VeraGrid Python API to **minimize transmission losses** via topology switching while **preserving (and improving) small-signal stability** through converter control tuning. Submissions are scored automatically across multiple operating scenarios.

Date 26 November 2026

Time 09:00–17:00

Venue Sala InnoEnergy, ETSEIB, Barcelona

Language English

Programme

Time	Session and milestone
09:00–09:15	**Hackathon kick-off:** Challenge presentation, stability criteria, and VeraGrid platform introduction.
09:15–10:15	**Ungraded warm-up: Reconstruct the function.** From matrices S and Φ, build expressions and a Python evaluator; validate on reference cases.
10:15–11:00	**Part 2: Static grid and topology switching.** Baseline evaluation, topology and dispatch optimisation with VeraGrid.
11:00–11:30	*Coffee break.*
11:30–12:00	**Hackathon support:** Dedicated mentoring and technical Q&A with TenSyGrid consortium researchers.
12:00–13:00	**Part 3: Dynamic layer and converter control tuning.** Computing eigenvalues, modal analysis, and stabilizing critical oscillation modes.
13:00–14:00	*Lunch break.*
14:00–15:30	**Part 4: Automation and batch resilience.** Multi-scenario batch sweep across 300 test operating points and parameter optimization.
15:30–16:00	*Coffee break and code freeze.* Submission of final scripts to the automated scoring harness.
16:00–17:00	**Hackathon wrap-up and winners announcement:** Team presentations, live leaderboard reveal, and awards ceremony.